Notation and conventions

This card is a translation layer between the assigned readings. It fixes the notation used in this course; it does not replace the derivations in the Week 1 reading route, Week 4 reading route, Week 5 reading route or Week 7 reading route. Use `:=` when introducing a definition and `=` for an asserted equality. Define domains, dimensions, supports and regularity conditions whenever they affect a calculation.

Sets, logic and functions

The real numbers are $\mathbb{R}$, Euclidean n-space is $\mathbb{R}^n$, the natural numbers are $\mathbb{N}$, and the empty set is $\emptyset$. If whether $0 \in \mathbb{N}$ matters, state the convention; for sample sizes and finite sums below, n is a positive integer. For sets, use $a \in A$ (membership), $A \subseteq B$ (inclusion), $A \cup B$ (union), $A \cap B$ (intersection) and A^c (complement relative to the stated universe). For propositions P and Q, use $\forall$ (for all), $\exists$ (there exists), $P \Rightarrow Q$ (implication) and $P \Leftrightarrow Q$ (equivalence). An implication is not an equality.

Write a function with its domain and codomain when they matter, for example

$$f : \mathbb{R}^n \to \mathbb{R}^m, \qquad \mathbf{x} \mapsto f(\mathbf{x}).$$

Here n is the number of inputs and m the number of outputs. Intervals use (a, b), $[a, b]$, $(a, b]$ or $[a, b)$; the brackets, not surrounding prose, state endpoint inclusion.

Logs, sums and products

The unqualified logarithm is natural: $\log x$ means base e, where e is Euler's number, and requires $x > 0$. For another base write $\log_b x$, where $b > 0$ and $b \neq 1$. In particular, log identities are used only where their arguments are in the logarithm's domain.

For scalars a_i indexed by $i = 1, \ldots, n$, write $\sum_{i=1}^{n} a_i$ and $\prod_{i=1}^{n} a_i$. A geometric sequence with first term a, common ratio r, and final index $N \in \mathbb{N} \cup \{0\}$ has

$$\sum_{t=0}^{N} ar^t = \begin{cases} a\dfrac{1 - r^{N+1}}{1 - r}, & r \neq 1, \\ (N+1)a, & r = 1. \end{cases}$$

The infinite version is written only after its convergence condition is checked:

$$\sum_{t=0}^{\infty} ar^t = \frac{a}{1 - r} \qquad \text{when } |r| < 1.$$

Scalars, vectors and matrices

Scalars use italic lower-case letters such as x, a and θ. Vectors use bold lower-case letters and are always columns; thus $\mathbf{x} \in \mathbb{R}^n$ means an $n \times 1$ vector. Matrices use bold upper-case letters. If $\mathbf{A} \in \mathbb{R}^{m \times n}$, it has m rows and n columns. The identity matrix is $\mathbf{I}_n$, and a zero matrix is $\mathbf{0}_{m \times n}$ when its dimensions need to be made explicit. Use $\mathbf{A}^\mathsf{T}$ and $\operatorname{rank}(\mathbf{A})$ for any rectangular matrix. For a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, use $\det(\mathbf{A})$ and $\operatorname{tr}(\mathbf{A})$; use $\mathbf{A}^{-1}$ only when that square matrix is nonsingular.

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the Euclidean inner product is $\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{x}^\mathsf{T}\mathbf{y}$. A linear system is $\mathbf{A}\mathbf{x} = \mathbf{b}$, with $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{b} \in \mathbb{R}^m$. For $\mathbf{A} \in \mathbb{R}^{n \times n}$, an eigenvalue $\lambda \in \mathbb{R}$ and corresponding eigenvector $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ satisfy $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$.

Differentiation and dimension checks

For a one-variable function use $f'(x)$, $f''(x)$, or $\mathrm{d}f/\mathrm{d}x$. For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is the **column**

$$\nabla f(\mathbf{x}) := \begin{bmatrix} \partial f/\partial x_1 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix} \in \mathbb{R}^n.$$

For a vector-valued map $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$ with components $f_1, \dots, f_m$, the Jacobian has one output per row and one input per column:

$$\mathbf{J}_{\mathbf{f}}(\mathbf{x}) := \left[\frac{\partial f_i}{\partial x_j}\right]_{i=1,\dots,m;\, j=1,\dots,n} \in \mathbb{R}^{m\times n}.$$

Thus the $1\times n$ Jacobian of a scalar function is the transpose of its $n\times 1$ gradient. A source row such as $[f_{x_1}, \dots, f_{x_n}]$ is therefore translated as $\nabla f = [f_{x_1}, \dots, f_{x_n}]^{\mathsf{T}}$.

The Hessian of f is the $n \times n$ matrix

$$\mathbf{H}_f(\mathbf{x}) := \left[\frac{\partial^2 f}{\partial x_i \partial x_j}\right]_{i,j=1,\dots,n}.$$

If the second partial derivatives are continuous near the point, the mixed partials agree and this Hessian is symmetric. For a small column change $\mathrm{d}\mathbf{x} \in \mathbb{R}^n$, the total differential is

$$\mathrm{d}f = \nabla f(\mathbf{x})^{\mathsf{T}} \mathrm{d}\mathbf{x}, \qquad (1 \times n)(n \times 1) = 1 \times 1.$$

For differentiable maps $\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^n$ and $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$, let $\mathbf{h} := \mathbf{f} \circ \mathbf{g}$. Then

$$\mathbf{J}_{\mathbf{h}}(\mathbf{u}) = \mathbf{J}_{\mathbf{f}}(\mathbf{g}(\mathbf{u}))\mathbf{J}_{\mathbf{g}}(\mathbf{u}), \qquad (m \times n)(n \times p) = m \times p.$$

If the outer map is scalar-valued, the equivalent column-gradient form is

$$\nabla(f \circ \mathbf{g})(\mathbf{u}) = \mathbf{J}_{\mathbf{g}}(\mathbf{u})^{\mathsf{T}} \nabla f(\mathbf{g}(\mathbf{u})).$$

For a differentiable path $\mathbf{x}(t)$ from a scalar t, this reduces to $\mathrm{d}f(\mathbf{x}(t))/\mathrm{d}t = \nabla f(\mathbf{x}(t))^{\mathsf{T}} \mathbf{x}'(t)$.

Suppose $F(x_0, \theta_0) = 0$, F is continuously differentiable near (x_0, θ_0), and $F_x(x_0, \theta_0) \neq 0$. Then the equation defines a differentiable local solution $x(\theta)$, and along that solution

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -\frac{F_\theta}{F_x}.$$

For a system $\mathbf{F} : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$, suppose $\mathbf{F}(\mathbf{x}_0, \theta_0) = \mathbf{0}$, the map is continuously differentiable nearby, and the Jacobian with respect to $\mathbf{x}$ is nonsingular at $(\mathbf{x}_0, \theta_0)$. The system then defines a differentiable local solution $\mathbf{x}(\theta)$. Let $\mathbf{J}_{\mathbf{F},\mathbf{x}}$ be the $n \times n$ Jacobian with respect to $\mathbf{x}$, and let $\mathbf{F}_\theta \in \mathbb{R}^n$ collect the partial derivatives with respect to θ. Along the local solution, the derivative is

$$\mathbf{x}'(\theta) = -\mathbf{J}_{\mathbf{F},\mathbf{x}}^{-1} \mathbf{F}_\theta.$$

If f is twice continuously differentiable near the scalar point a, Taylor notation records its centre: the second-order approximation about a is

$$f(x) \approx f(a) + f'(a)(x-a) + \tfrac{1}{2} f''(a)(x-a)^2.$$

Curvature and optimisation conventions

A quadratic form is $q(\mathbf{x}) := \mathbf{x}^{\mathsf{T}} \mathbf{A} \mathbf{x}$, where $\mathbf{A} \in \mathbb{R}^{n\times n}$. Definiteness tests use the symmetric part $\mathbf{A}_s := \frac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{T}})$. After the terms are defined, $\mathbf{A}_s \succ 0$, $\mathbf{A}_s \succeq 0$, $\mathbf{A}_s \prec 0$ and $\mathbf{A}_s \preceq 0$ mean positive definite, positive semidefinite,

negative definite and negative semidefinite. Curvature conclusions are global only with the relevant domain and differentiability assumptions. Quasiconcavity means convex upper contour sets; it is not a synonym for concavity.

For a scalar maximisation problem with equality constraints $h_k(\mathbf{x}, \mathbf{a}) = 0$, $k = 1, \dots, K$, where K is the number of equalities and $\mathbf{a}$ is a parameter vector, use

$$\mathcal{L}(\mathbf{x}, \mu; \mathbf{a}) := f(\mathbf{x}, \mathbf{a}) + \sum_{k=1}^{K} \mu_k h_k(\mathbf{x}, \mathbf{a}).$$

For continuously differentiable data, at a regular local optimum where the equality-constraint gradients are linearly independent, unrestricted multipliers μ_k exist such that stationarity in $\mathbf{x}$ and all equalities hold. Solutions of these equations are candidates; concavity, feasibility and other stated assumptions are needed for a global conclusion. The sign of a multiplier is tied to the constraint's encoding. For a value function $V(\mathbf{a})$, the envelope derivative is

$$\frac{\partial V}{\partial a_r} = \left.\frac{\partial \mathcal{L}}{\partial a_r}\right|_{*}$$

under the stated regularity and optimality conditions, where $*$ denotes the optimum.

The course-wide Kuhn–Tucker convention is **maximise** $f(\mathbf{x})$ subject to $g_j(\mathbf{x}) \geq 0$, $j = 1, \dots, J$, and $h_k(\mathbf{x}) = 0$, $k = 1, \dots, K$. With non-negative multipliers $\lambda_j \geq 0$ and unrestricted equality multipliers μ_k, write

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) := f(\mathbf{x}) + \sum_{j=1}^{J} \lambda_j g_j(\mathbf{x}) + \sum_{k=1}^{K} \mu_k h_k(\mathbf{x}).$$

The four checks are stationarity $\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}$, primal feasibility $g_j \geq 0$, $h_k = 0$, dual feasibility $\lambda_j \geq 0$, and complementary slackness $\lambda_j g_j = 0$. Write a restriction such as $x_i \geq 0$ explicitly as a constraint. Rewrite a minimisation question into this maximisation convention before applying the template. With differentiable data and a suitable constraint qualification, these conditions are necessary at a regular local optimum. In general they identify candidates, not optima; they are sufficient only when an applicable concavity result and a convex feasible set supply the required global conditions.

Probability and random variables

Use Ω for the sample space and upper-case Roman letters such as A and B for events. Their probabilities are $\mathbb{P}(A)$, and conditional probability is $\mathbb{P}(A \mid B)$. Independence can be recorded as $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Random variables use upper-case letters X, Y; their realised values use lower-case letters x, y. Random vectors use bold upper-case letters.

For a discrete X, write its PMF as $p_X(x)$; for a continuous variable, write its density as $f_X(x)$; in either case write the support alongside the formula. The CDF is $F_X(x) := \mathbb{P}(X \leq x)$. Use $\mathbb{E}[X]$, $\mathrm{Var}(X)$, $\mathrm{Cov}(X, Y)$ and $\mathrm{Corr}(X, Y)$. Conditional expectation is $\mathbb{E}[X \mid Y]$ as a random variable and $\mathbb{E}[X \mid Y = y]$ at the value y.

When conditional probability is written as a ratio, $\mathbb{P}(A \mid B) := \mathbb{P}(A \cap B)/\mathbb{P}(B)$ requires $\mathbb{P}(B) > 0$.

The following parameterisations are the course standard. In the table, p is a success probability, n is a positive integer, λ is a rate or Poisson mean, μ is a normal mean, σ^2 is a variance, and ν is a positive degrees-of-freedom value.

Distribution	Notation, support and parameterisation
Bernoulli	$X \sim \text{Bernoulli}(p)$, $0 \leq p \leq 1$, support $\{0, 1\}$, and $\mathbb{P}(X = 1) = p$.
Binomial	$X \sim \text{Binomial}(n, p)$, $0 \leq p \leq 1$, the number of successes in n independent Bernoulli trials; support $\{0, \dots, n\}$.
Geometric	$X \sim \text{Geometric}(p)$, $0 < p \leq 1$, number of trials through the first success; support $\{1, 2, \dots\}$.

Distribution	Notation, support and parameterisation
Poisson	$X \sim \text{Poisson}(\lambda)$, $\lambda > 0$ is both rate and mean; support $\{0, 1, ...\}$.
Uniform	$X \sim \text{Uniform}(a, b)$, $a < b$, density $1/(b-a)$ on $[a, b]$.
Exponential	$X \sim \text{Exponential}(\lambda)$, rate $\lambda > 0$, density $\lambda e^{-\lambda x}$ for $x \geq 0$, mean $1/\lambda$.
Normal	$X \sim \mathcal{N}(\mu, \sigma^2)$, $\sigma^2 > 0$, support $\mathbb{R}$; the second parameter is the variance.
Student's t	$T \sim t_\nu$, the standard t distribution with ν degrees of freedom and support $\mathbb{R}$.
Chi-squared	$Q \sim \chi^2_\nu$, the chi-squared distribution with ν degrees of freedom and support $[0, \infty)$; use it only where the scope permits.

Sampling and inference distinctions

A random sample from a population distribution F_θ indexed by the parameter θ is written

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} F_\theta.$$

The label `iid` asserts both independence and identical distribution; do not use it for independent but non-identically distributed observations. For $n \geq 2$, use

$$\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

After observing data $\mathbf{x} = (x_1, \dots, x_n)$, write the corresponding values as $\bar{x}$ and s^2.

The population **parameter** θ is fixed but unknown. An **estimand** is the population quantity of interest, possibly a function of the parameter. An **estimator** is a random variable, for example $\hat{\theta} = T(X_1, \dots, X_n)$; its observed numerical value $T(\mathbf{x})$ is the **estimate**. Bias and variance refer to the estimator's repeated-sampling distribution, not to one estimate.

For observed data $\mathbf{x}$, define the likelihood $L(\theta; \mathbf{x})$ on a stated parameter space Θ. The log-likelihood is $\ell(\theta; \mathbf{x}) := \log L(\theta; \mathbf{x})$, and an MLE is a maximiser over $\theta \in \Theta$, including boundary checks when relevant.

Use H_0 and H_1 for null and alternative hypotheses. The significance level α is the Type I error probability under H_0; at a specified alternative, $\beta(\theta)$ is the Type II error probability and $1 - \beta(\theta)$ is power. A p-value is, under H_0, the probability of a result at least as incompatible with H_0 as the observed result, using the stated statistic and tail convention. It is not the probability that H_0 is true. A confidence interval must name its confidence level and method; write the random interval as $[L(\mathbf{X}), U(\mathbf{X})]$ and report numerical endpoints after observing data.

Final checks

- Domains, endpoint conventions, dimensions, parameter restrictions and distribution supports are visible wherever they affect validity.
- Vectors and gradients are columns; a Jacobian of $\mathbb{R}^n \to \mathbb{R}^m$ is $m \times n$; chain-rule factors multiply in that order.
- Infinite sums carry a convergence condition, and `log` means the natural logarithm unless a base is written.
- Equality multipliers follow the displayed constraint encoding; KKT uses maximisation, $g_j \geq 0$, $\lambda_j \geq 0$, feasibility and complementary slackness.
- Upper-case random variables are separated from lower-case observations; `iid` means independent and identically distributed; S^2 uses $n - 1$.
- A p-value is a tail probability under H_0, and every confidence interval states its level and method.